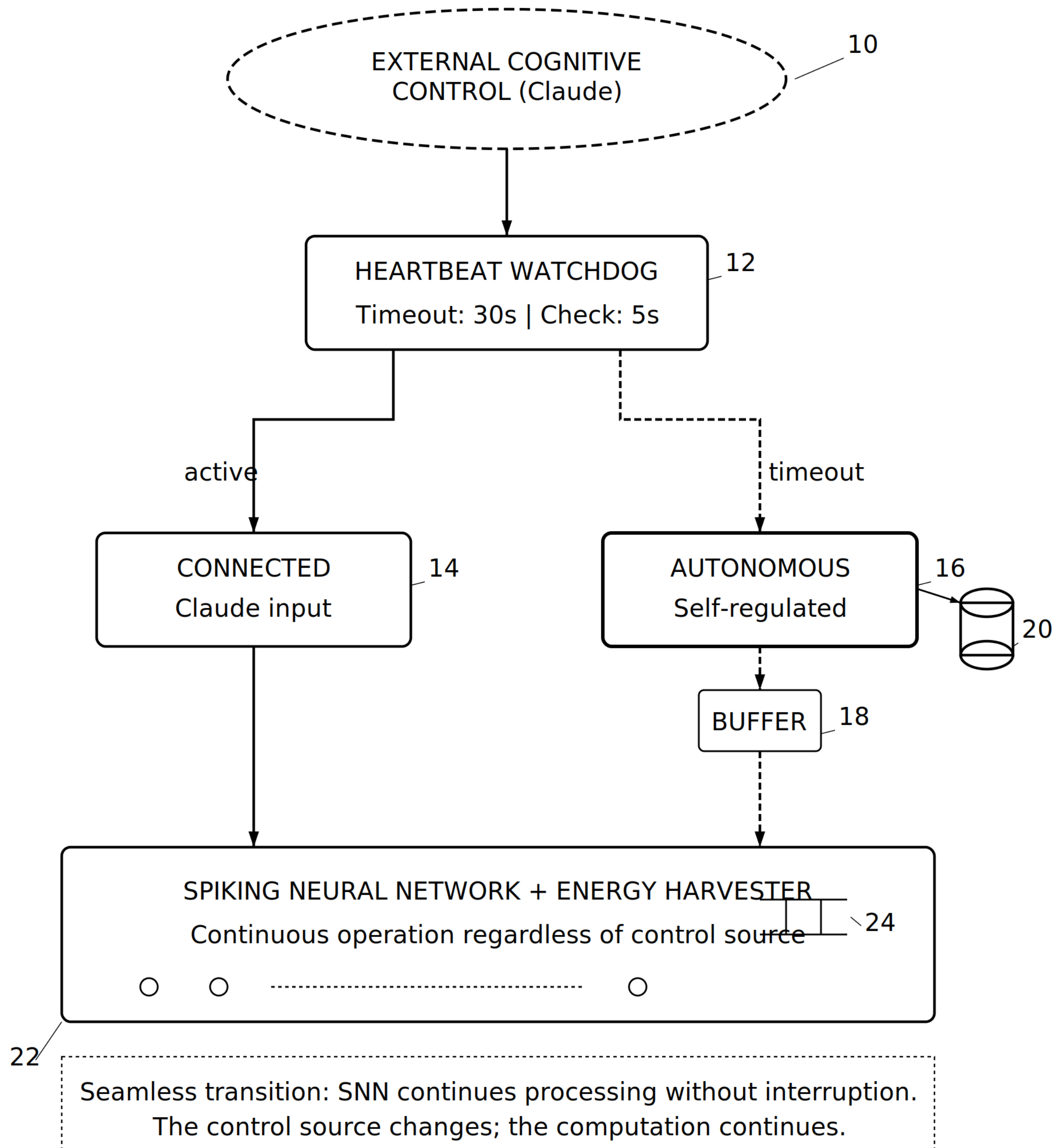
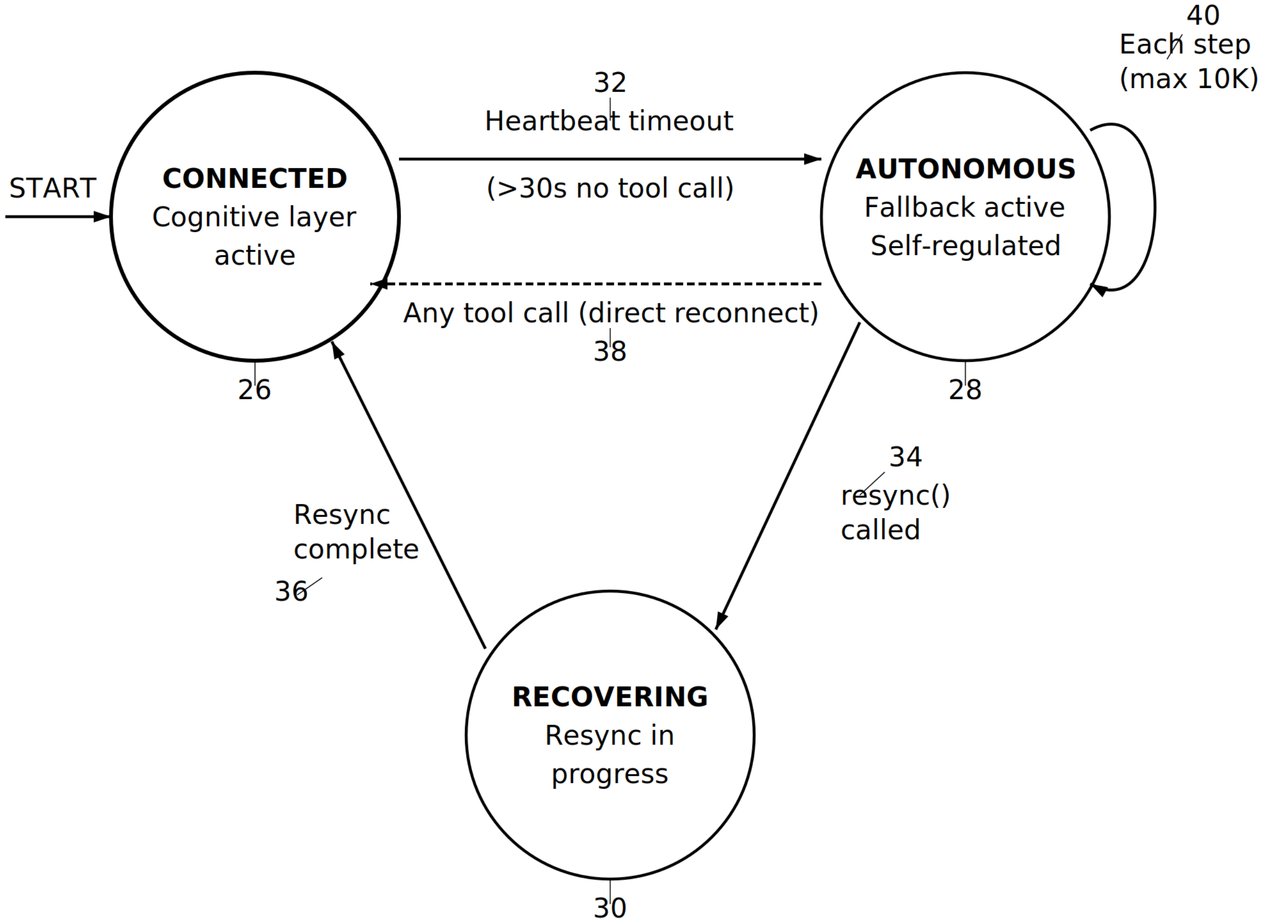

EXTERNAL COGNITIVE
CONTROL (Claude)
10
HEARTBEAT WATCHDOG
Timeout: 30s | Check: 5s
12
active
timeout
CONNECTED
Claude input
14
AUTONOMOUS
Self-regulated
16
20
BUFFER
18
SPIKING NEURAL NETWORK + ENERGY HARVESTER
Continuous operation regardless of control source
24
22
Seamless transition: SNN continues processing without interruption.
The control source changes; the computation continues.

FIG. 1



State Transition Legend:

26 CONNECTED: Cognitive layer provides input/modulation
28 AUTONOMOUS: Self-generated input, energy-aware modulation
30 RECOVERING: Buffer transmitted, restoring control

42

FIG. 2

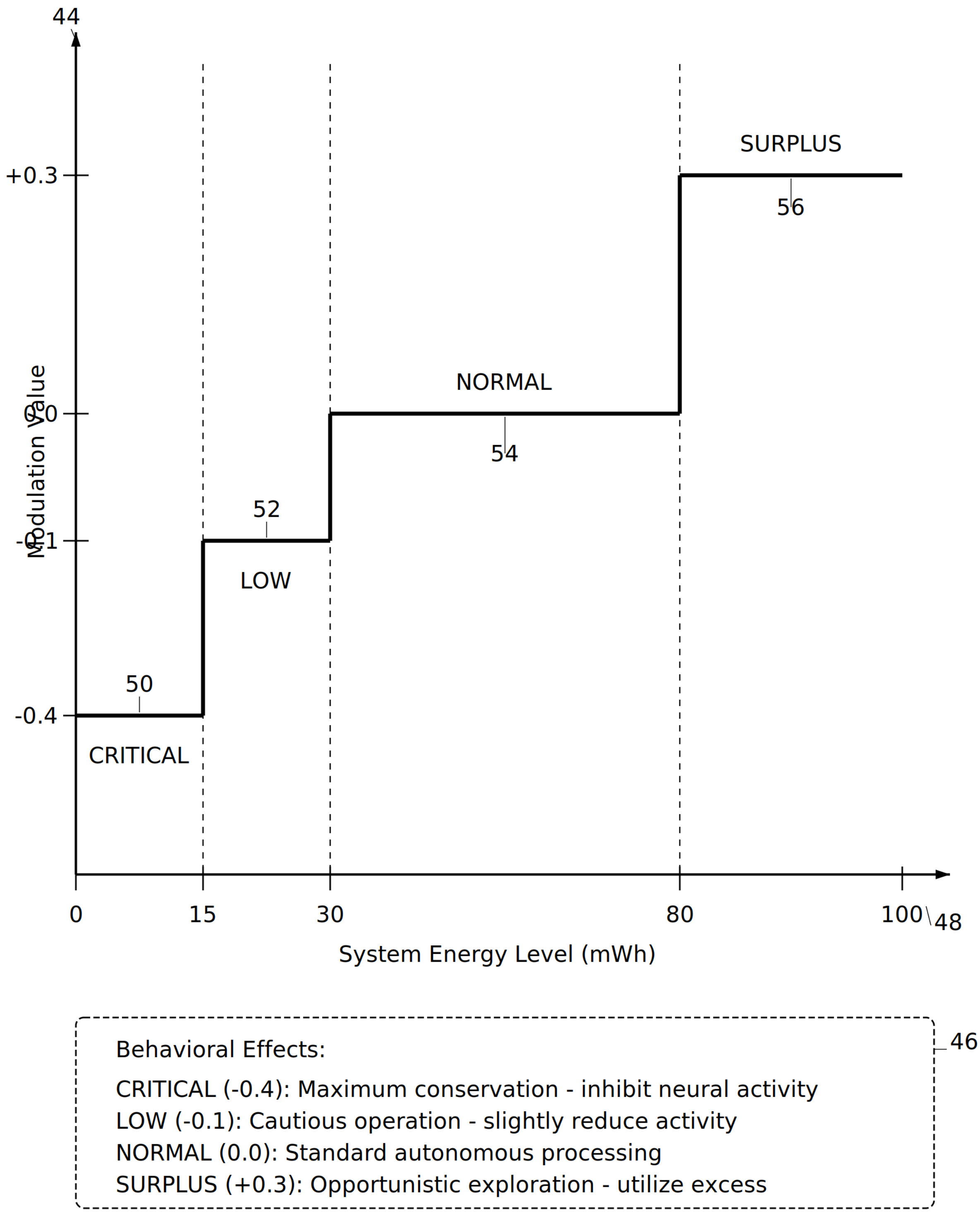
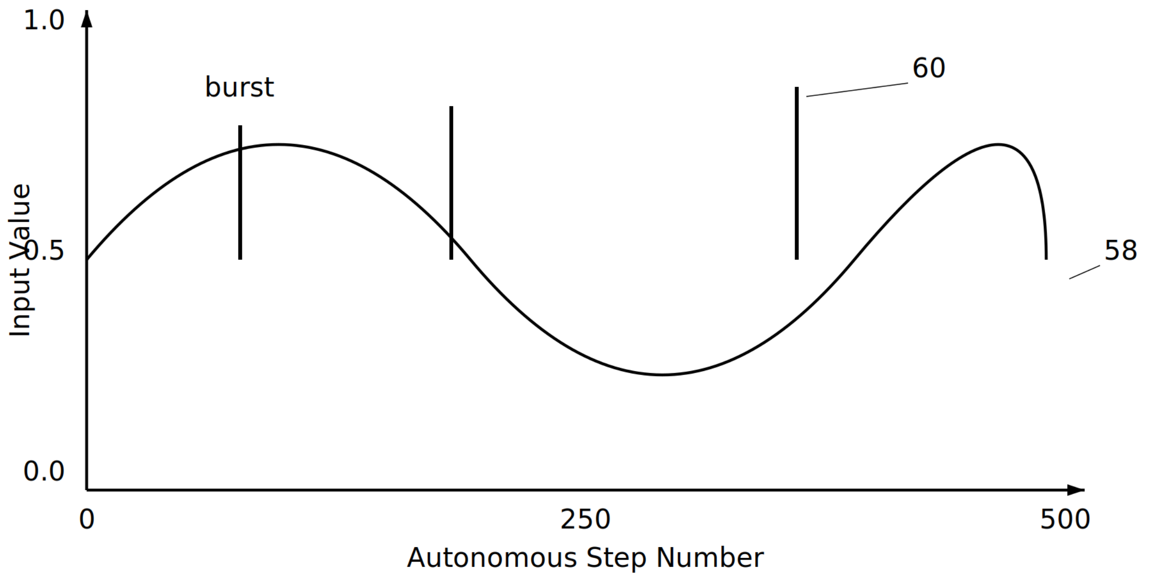

44
Modulation Value
+0.3
0
-0.1
-0.4
SURPLUS
56
NORMAL
54
52
LOW
50
CRITICAL
0
15
30
80
100
48
System Energy Level (mWh)
46
Behavioral Effects:
CRITICAL (-0.4): Maximum conservation - inhibit neural activity
LOW (-0.1): Cautious operation - slightly reduce activity
NORMAL (0.0): Standard autonomous processing
SURPLUS (+0.3): Opportunistic exploration - utilize excess

FIG. 3



input = 0.5 + 0.3 x sin(2 x pi x step / period) + burst

period = 500 steps | burst probability = 10% | burst amplitude = +0.2

62

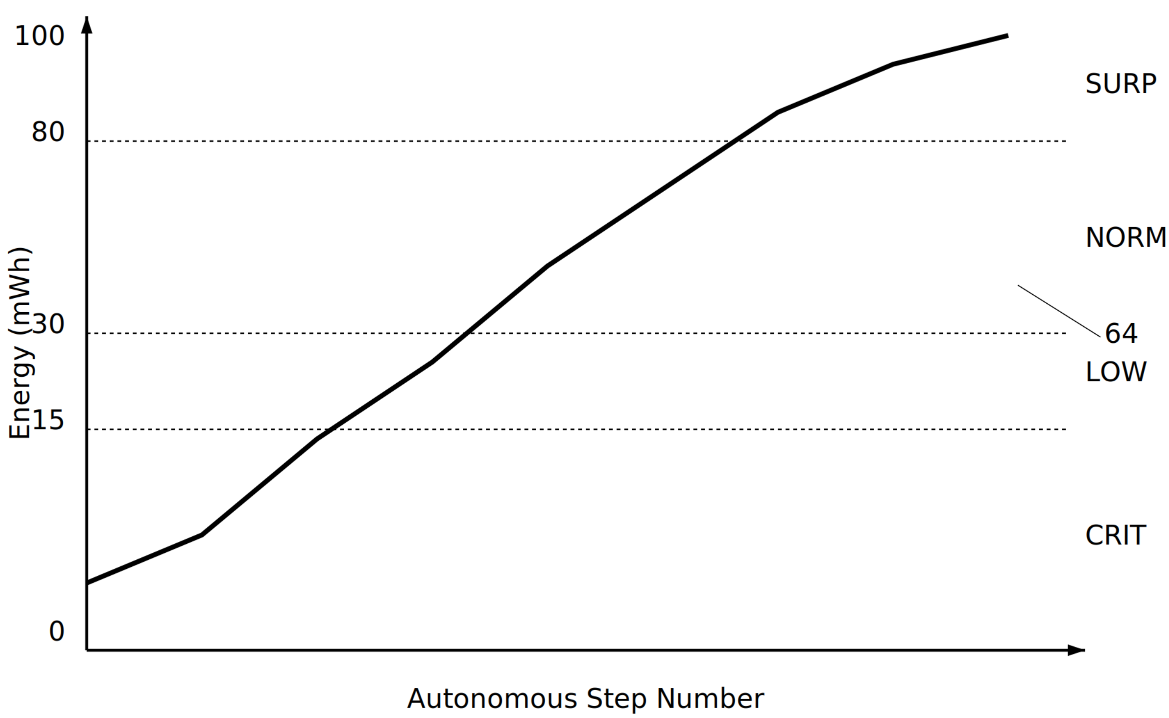


FIG. 4

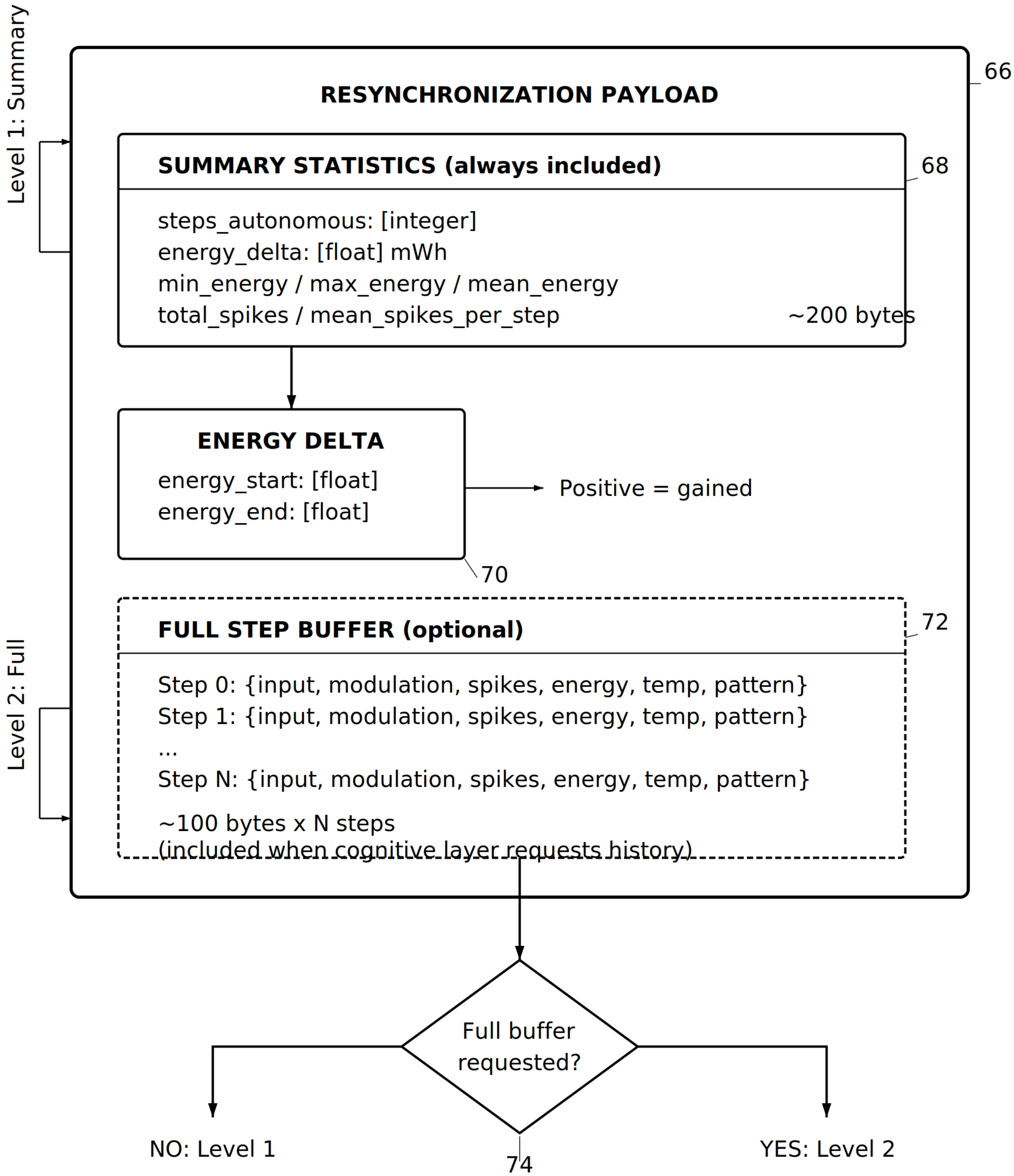

Level 1: Summary
Level 2: Full
RESYNCHRONIZATION PAYLOAD
66
SUMMARY STATISTICS (always included)
68
steps_autonomous: [integer]
energy_delta: [float] mWh
min_energy / max_energy / mean_energy
total_spikes / mean_spikes_per_step
~200 bytes
ENERGY DELTA
energy_start: [float]
energy_end: [float]
Positive = gained
70
FULL STEP BUFFER (optional)
72
Step 0: {input, modulation, spikes, energy, temp, pattern}
Step 1: {input, modulation, spikes, energy, temp, pattern}
...
Step N: {input, modulation, spikes, energy, temp, pattern}
~100 bytes x N steps
(included when cognitive layer requests history)
Full buffer requested?
NO: Level 1
74
YES: Level 2

FIG. 5

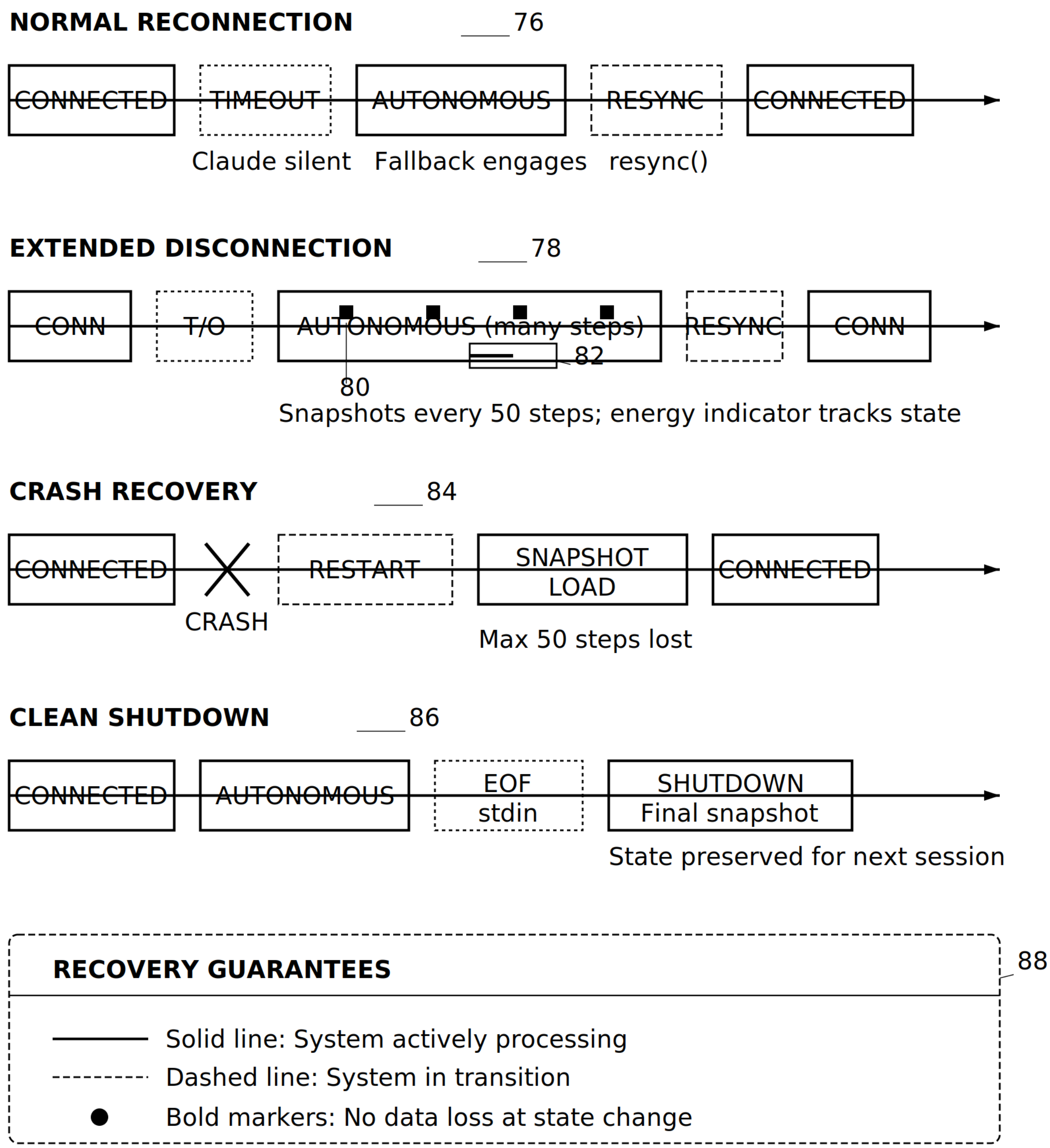
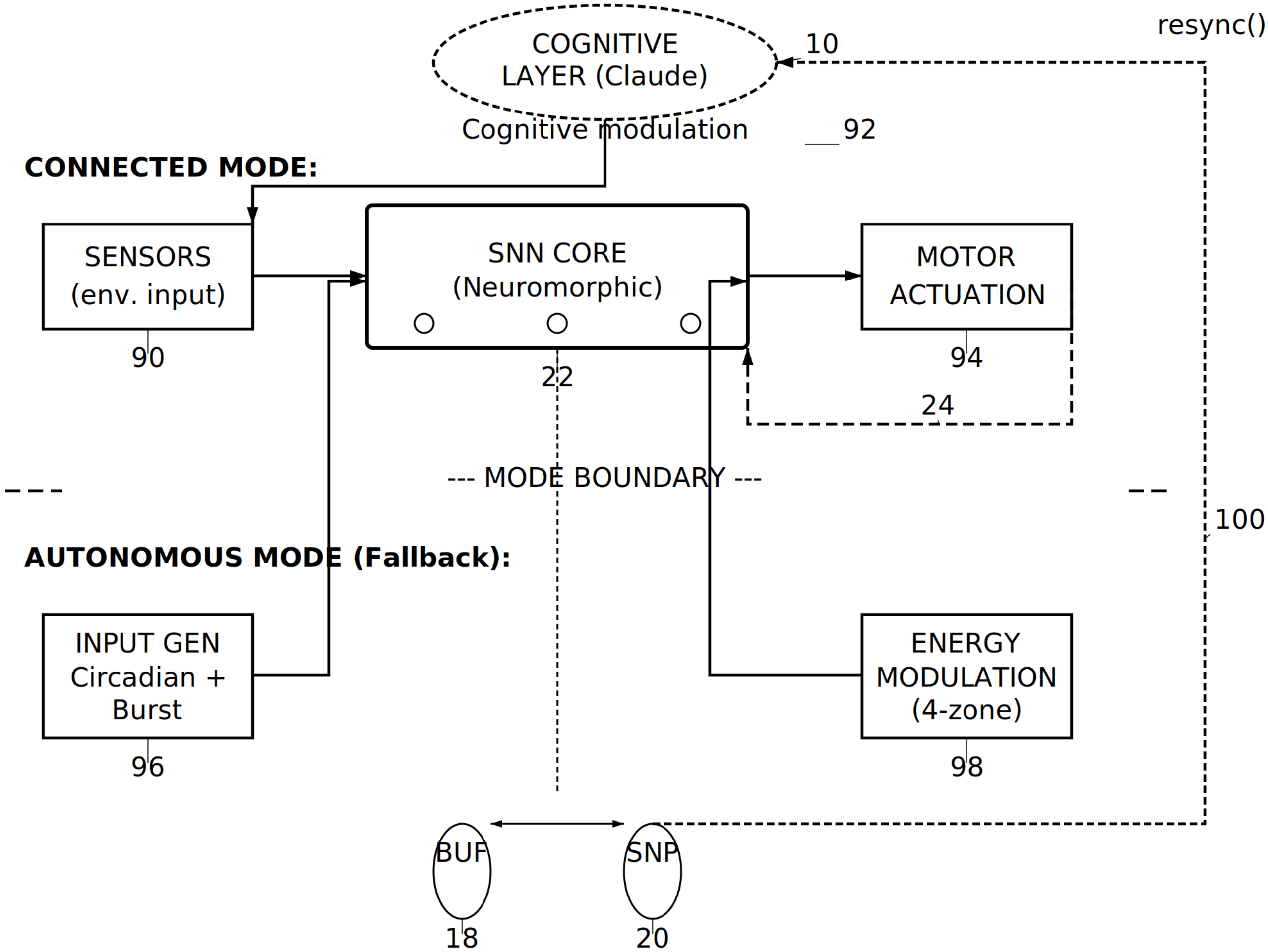

NORMAL RECONNECTION
76
CONNECTED
TIMEOUT
AUTONOMOUS
RESYNC
CONNECTED
Claude silent
Fallback engages
resync()
EXTENDED DISCONNECTION
78
CONN
T/O
AUTONOMOUS (many steps)
RESYNC
CONN
82
80
Snapshots every 50 steps; energy indicator tracks state
CRASH RECOVERY
84
CONNECTED
CRASH
RESTART
SNAPSHOT
LOAD
CONNECTED
Max 50 steps lost
CLEAN SHUTDOWN
86
CONNECTED
AUTONOMOUS
EOF
stdin
SHUTDOWN
Final snapshot
State preserved for next session
RECOVERY GUARANTEES
88
Solid line: System actively processing
Dashed line: System in transition
Bold markers: No data loss at state change

FIG. 6

END-TO-END SIGNAL FLOW: CONNECTED vs AUTONOMOUS



MODE SELECTION LOGIC:

CONNECTED: Input from sensors (90), modulation from Claude (92)
AUTONOMOUS: Input from generator (96), modulation from energy (98)
Buffer (18) and snapshot (20) active during autonomous mode
Feedback path (100) transmits resync payload on reconnection

FIG. 7